



MAARULA CLASSES

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**NIMCET ORIGINAL
PAPER 2023
PYP No: 16**



SCAN QR OF OUR APP MAARULA MATHEM

COMPUTER AWARENESS

- The time required for fetching and execution of one machine instruction is
(a) Delay time (b) Seek time (c) Real time (d) CPU cycle
- The number of minterms in a n variable truth table is
A. n^2 B. $(n-1)^2$ C. 2^n D. 2^{n-1}
(a) D (b) B (c) A (d) C
- The maximum and minimum value represented in signed 16-bit 2's complement representation are
(a) 0 and 32767 (b) 0 and 65535
(c) -16384 and 16383 (d) -32768 and 32767
- A bulb in the staircase has two switches, one switch is at the ground floor and the other one is at the first floor. The bulb can be turned ON and also can be turned OFF by any of the switches irrespective of the state of the other switch. The logic of the switching of the bulb resembles
(a) XOR gate (b) AND gate (c) OR gate (d) XNOR gate
- What is a potential problem of 1's complement representation of numbers?
(a) Binary subtractions are not possible
(b) There are two different representations of zero
(c) Multiplication of two numbers cannot be carried out
(d) Binary additions are not possible
- A wrong sentence related to FAT 32 and NTFS file system is
(a) FAT 32 has lower disk utilization compared to NTFS
(b) Read and write speeds of NTFS are faster than that of FAT 32
(c) FAT 32 store individual files of size up to 32 GB
(d) NTFS stands for New Technology File System
- Consider the circuit shown below and find minimum number of NAND gates required to design it. a

 (a) 4 (b) 6 (c) 3 (d) 5
- Consider the following minterm expression for $F(F, P, Q, R, S) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$. The minterms 2, 7, 8 and 13 are don't care terms. The minimum sum of products form for F is
A. $\bar{Q}S + QS$ B. $\bar{Q}S + QS$
C. $\bar{Q}R\bar{S} + \bar{Q}RS + Q\bar{R}S + QRS$ D. $\bar{P}Q\bar{S} + \bar{P}QS + PQ\bar{S} + PQS$
(a) A (b) D (c) B (d) C
- The reduced form of the Boolean Function $F = xyz + xyz' + x'yz + xy'z$ is
(a) $xy + yz$ (b) $x + yz + xz$ (c) $x + y + z$ (d) $xy + yz + xz$
- If a processor clock is rated as 2500 million cycles per second, then its clock period is
A. 2.50×10^{-10} sec B. 4.00×10^{-10} sec
C. 1.00×10^{-10} sec D. 5.00×10^{-10} sec
(a) D (b) C (c) A (d) B
- Suppose we have a 10-bit computer that uses 10-bit int (2's complement representation). The number representation of -35 is
A. 0000100011 B. 1100100011
C. 1111011101 D. 1111011101
(a) B (b) D (c) A (d) C
- Consider the following Boolean expression for $F(F, P, Q, R, S) = PQ + \bar{P}QR + \bar{P}Q\bar{R}S$. The minimum sum of products form of F is
A. $PQ + QR + QS$ B. $P + Q + R + S$
C. $\bar{P} + \bar{Q} + \bar{R} + \bar{S}$ D. $\bar{P}R + \bar{P}\bar{R}S + P$
(a) D (b) A (c) C (d) B
- What is the name of the storage device that compensates the difference in rates of flow of data from one device to another?
(a) Cache (b) Buffer (c) Concentrator (d) RAM

- Equivalent of the decimal number $(25.375)_{10}$ in binary form
A. $(11001.011)_2$ B. $(11101.011)_2$
C. $(11011.111)_2$ D. $(11001.101)_2$
(a) A (b) D (c) B (d) C
- Which of the following registers is used to keep track of address of the memory location where the next instruction is located?
(a) Program Counter (b) Memory Data Register
(c) Instruction counters (d) Memory Address Register
- A CPU generates 32-bit virtual addresses. The page size is 4KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is
(a) 13 bits (b) 20 bits (c) 11 bits (d) 15 bits
- Which of the following is true about Von Neumann architecture?
(a) It has separate storage for input/output operations
(b) It has a separate processing unit for data and instructions
(c) It has separate memory for data and instructions
(d) It has a single memory unit for both data and instructions
- Let $\oplus$ and $\odot$ denote the Exclusive-OR and Exclusive-NOR operations respectively. Which one of the following is not correct?
A. $\bar{P} \oplus \bar{Q} = P \odot Q$ B. $\bar{P} \oplus Q = P \odot Q$
C. $\bar{P} \oplus \bar{Q} = P \oplus Q$ D. $(P \oplus \bar{P}) \oplus Q = (P \odot \bar{P}) \odot \bar{Q}$
(a) C (b) D (c) B (d) A
- Suppose we have a 10-bit computer that uses 10-bit floating-point computational unit (Float number uses IEEE floating-point arithmetic where a floating point number has 1 sign bit, 5 exponent bits, and 4 fraction bits). The representation for $+\infty$ (plus infinity) is
(a) 0111110000 (b) 1111110000
(c) 0000001111 (d) 011111 1111
- How many 32 K $\times$ 1 RAM chips are needed to provide a memory Capacity of 256 K-bytes?
(a) 8 (b) 128 (c) 64 (d) 32

ANSWER

1.	d	2.	d	3.	d	4.	a	5.	b
6.	c	7.	a	8.	c	9.	d	10.	d
11.	d	12.	b	13.	b	14.	a	15.	a
16.	d	17.	d	18.	b	19.	a	20.	c

SOLUTION

- (d)** The time required for fetching and executing one machine instruction is typically measured in terms of CPU cycles. A CPU cycle, also known as a clock cycle, is the basic unit of time used by a processor to execute instructions. Each instruction requires a certain number of CPU cycles to complete, which includes the time for fetching the instruction from memory, decoding it, and executing the corresponding operation.
- (d)** The number of minterms in a n -variable truth table is 2^n .
- (d)** In a signed 16-bit 2's complement representation, the most significant bit (MSB) is reserved for the sign of the number. The remaining 15 bits represent the magnitude of the number. For a signed 16-bit representation, the range of values that can be represented is from -2^{15} to $2^{15}-1$.
 $-2^{15} = -32768$
 $2^{15}-1 = 32767$

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Therefore, the maximum value represented is -32768, and the minimum value represented is 32767.

4. (a) In this scenario, the behaviour of the switches can be represented by an XOR gate. An XOR gate, also known as an Exclusive OR gate, produces an output of "true" (or ON) only when the number of input signals that are true is odd. In this case, the bulb will be ON when either switch is flipped, regardless of the state of the other switch.
5. (b) In 1's complement representation of numbers, the sign of a number is determined by the leftmost bit (MSB), where 0 represents a positive number and 1 represents a negative number. To obtain the 1's complement of a number, each bit of the original number is inverted. However, one potential problem with 1's complement representation is that there are two different representations of zero. In regular binary representation, zero is represented as all zeros. However, in 1's complement representation, zero can be represented as either all zeros or all ones.
6. (c) In general, NTFS (New Technology File System) tends to provide faster read and write speeds compared to FAT32 (File Allocation Table). NTFS incorporates various improvements over FAT32, including better file organisation, enhanced security features, and improved performance.
7. (a) A NAND gate is a type of logic gate that performs the logical operation called NAND (NOT-AND). It is a combination of an AND gate followed by a NOT gate. The minimum number of NAND gates required to design given circuit is 4.
8. (c) $F(P, Q, R, S) = \Sigma(0, 2, 5, 7, 8, 10, 13, 15)$

RS \ PQ	00	01	11	10
00	1			X
01		1	X	
11		X	1	
10	X			1

While putting the terms to K-map following things happen,

- 3rd and 4th columns are swapped
- 3rd and 4th rows,
- term 2 is going to (0,3) column instead of (0,2)
- 8 is going to (3,0) instead of (2,0)

Solving, the above K-map, we get $\bar{Q}\bar{S} + QS$

9. (d) Let's simplify the Boolean expression

$$F = xyz + xyz' + x'yz + xy'z.$$

$$xy + yz$$

We can factor out y from the first two terms: $y(xz + z')$. The expression $xy + yz$ is not equivalent to F .

$$x + yz + xz$$

We can factor out z from the first and last terms: $xz(y + x' + 1)$.

The expression $x + yz + xz$ is not equivalent to F .

$$x + y + z$$

This expression is not equivalent to F as it does not contain the term $xy'z$.

$$xy + yz + xz$$

We can see that this expression matches the given Boolean function F . It is the reduced form of F .

Therefore, the correct reduced form of

$$F = xyz + xyz' + x'yz + xy'z \text{ is } xy + yz + xz.$$

10. (d) To find the clock period of a processor given its clock rate, we can use the formula:

$$\text{Clock Period} = \frac{1}{\text{Clock Rate}}$$

In this case, the processor clock rate is 2500 million cycles per second, or 2500MHz.

Converting MHz to Hz :

$$2500\text{MHz} = 2500 \times 10^6 \text{ Hz}$$

Now, we can calculate the clock period

$$\text{Clock Period} = \frac{1}{(2500 \times 10^6 \text{ Hz})}$$

Clock Period = 0.0000004 seconds or 400 nanoseconds
Therefore, the clock period of the processor is 400 nanoseconds or $4.00 \times 10^{-10} \text{ sec}$.

11. (d) To represent -35 in a 10-bit computer that uses 10-bit two's complement representation, follow these steps:

Represent the positive value of 35 in binary: 0000100011. Invert all the bits: 1111011100.

Add 1 to the inverted value: 1111011101.

Thus, the 10-bit two's complement representation of -35 would be 1111011101.

12. (b)

$$F(P, Q, R, S) = PQ + P'QR + P'QR'S$$

$$= Q(P + P'R + P'R'S)$$

$$= Q(P + R + R'S)$$

$$= Q(P + R + S) = PQ + QR + QS$$

13. (b) A buffer is a storage device or area that is used to compensate for the difference in rates of flow of data between two devices. It temporarily holds data when there is a difference in data transfer rates or when the receiving device is not ready to process the data. Buffers help smooth out the data transfer process and prevent data loss or disruption.

14. (a) To convert the decimal number $(25.375)_{10}$ to binary form, we can convert the integer and fractional parts separately.

INTEGER PART

The integer part of $(25.375)_{10}$ is 25. To convert it to binary, we divide it by 2 repeatedly until the quotient becomes 0, and read the remainders in reverse order.

$$25 \div 2 = 12 \text{ remainder } 1$$

$$12 \div 2 = 6 \text{ remainder } 0$$

$$6 \div 2 = 3 \text{ remainder } 0$$

$$3 \div 2 = 1 \text{ remainder } 1$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Reading the remainders in reverse order, we get the binary representation of the integer part: $(25)_{10} = (11001)_2$.

Fractional Part

The fractional part of $(25.375)_{10}$ is 0.375. To convert it to binary, we multiply it by 2 repeatedly until the fractional part becomes 0, and read the integer parts.

$$0.375 \times 2 = 0.75 \rightarrow 0$$

$$0.75 \times 2 = 1.5 \rightarrow 1$$

$$0.5 \times 2 = 1.0 \rightarrow 1$$

Reading the integer parts, we get the binary representation of the fractional part: $(0.375)_{10} = (0.011)_2$. Combining the binary representation of the integer and fractional parts, we get:

$$(25.375)_{10} = (11001.011)_2 \text{ in binary form.}$$





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15. (a) The Program Counter (PC) is a register used in computer architecture to keep track of the address of the memory location where the next instruction to be fetched is located. It is also sometimes referred to as the Instruction Pointer (IP) in certain computer architectures. The PC is automatically incremented after each instruction is fetched, allowing the processor to fetch the next instruction from the sequential memory location.

16. (d) To determine the minimum size of the TLB tag, we need to consider the number of page table entries the TLB can hold and the page size.

Given:

CPU generates 32-bit virtual addresses. TLB can hold a total of 128 page table entries. The TLB is 4-way set associative.

Page size is 4 KB.

To calculate the TLB tag size, we can use the formula: Tag size = Address size - Index size - Offset size First, we need to find the index size. Since, the TLB is 4-way set associative and can hold 128 page table entries, the number of sets is $128/4 = 32$. Therefore, the index size is $\log_2(32) = 5$ bits.

Next, we calculate the offset size. The page size is 4 KB, which is equivalent to (2^{12}) bytes. So, the offset size is $\log_2(2^{12}) = 12$ bits.

Now, we can calculate the tag size: Tag size = 32 bits - 5 bits - 12 bits = 15 bits Therefore, the minimum size of the TLB tag is 15 bits.

17. (d) It has a single memory unit for both data and instructions. Von Neumann architecture, named after computer scientist John von Neumann, is a computer architecture design that is widely used in most modern computers. It is characterised by the following features:

It has a single memory unit: In Von-Neumann architecture, both instructions and data are stored in the same memory unit. This

means that the program instructions and data are fetched from and stored in the same memory.

It has a Central Processing Unit (CPU): The CPU is responsible for executing instructions and performing calculations. It fetches instructions from memory, decodes them, and executes them.

18. (b) XOR gate is a digital logic gate that gives a true (1 or HIGH) output when the number of true inputs is odd. The XNOR (Exclusive-NOR) gate is a combination XOR gate followed by an inverter. Its output is "true" if the inputs are the same, and "false" if the inputs are different.

19. (a) The correct representation for $+\infty$ (plus infinity) in the given 10-bit floating point format using IEEE floating-point arithmetic would be:

0111110000

In this representation, the sign bit is 0 (indicating a positive value), the exponent bits are all 1s (11111), and the fraction bits are all 0s (0000). This follows the IEEE 754 standard for representing infinity in floating-point arithmetic.

20. (c) Data

RAM chip's capacity = $32K \times 1 = 32 \times 1024 \times 1$ bits

Memory Capacity = $256K$ bytes = $256 \times 1024 \times 8$ bits

Formula

Number of chips needed

$$= \frac{\text{Memory Capacity}}{\text{RAM Chip's Capacity}}$$

Calculation

$$\text{Number of chips needed} = \frac{256 \times 1024 \times 8}{32 \times 1024 \times 1} = 64 \text{ chips}$$